Prefix Sums for Range Queries

Topic: CompetitiveProgramming File: c2.pdf

PROBLEM

Answer many queries asking for the sum of array elements from index L through index R, inclusive.

PREPROCESSING

Build prefix[0]=0 and prefix[i+1]=prefix[i]+A[i]. The sum of A[L..R] is prefix[R+1]-prefix[L].

EXAMPLE

A = [2,4,1,7], prefix = [0,2,6,7,14]. Query L=1, R=3 gives 14-2 = 12.

COMPLEXITY

Build time: O(n). Each range query: O(1). Extra space: O(n).

EDGE CASES

Be consistent about zero-based indexing and inclusive endpoints. Use a wide integer type if sums may overflow.